

Design Proposal: A Mach–Zehnder Interferometer for Group-Index Extraction on a Silicon-on-Insulator Platform

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Abstract—This report describes the design, modelling, mask layout, fabrication and measurement of a length-imbalanced Mach–Zehnder interferometer (MZI) suite on a silicon-on-insulator (SOI) platform. The objective is to provide an experimental platform for extracting the waveguide group index n_g of a $220\text{ nm} \times 500\text{ nm}$ strip waveguide from the free-spectral range (FSR) of the measured transmission spectrum. Four unbalanced MZIs with arm-length differences $\Delta L \in \{100, 150, 200, 250\}\text{ }\mu\text{m}$, plus a calibration loop-back structure, were designed, laid out and fabricated on the Applied Nanotools NanoSOI process. Two independent FSR extraction methods (dip-to-dip spacing and autocorrelation), applied inside the $1540\text{--}1560\text{ nm}$ fitting window recommended by the course, give group-index estimates in the range $4.92\text{--}6.41$ across the four devices, which is above the physically expected value of ~ 4.19 for this cross-section. All extracted values fall outside the nine-point corner-analysis window $n_g \in [4.156, 4.241]$ obtained from a $\pm 20\text{ nm}$ width sweep, and the strong monotonic dependence of the apparent n_g on the labelled ΔL suggests a mask-to-label mismatch that must be resolved before the corner window can be trusted as the sole diagnostic.

Index Terms—Silicon photonics, Mach–Zehnder interferometer, group index, free spectral range, strip waveguide, SiEPIC.

I. INTRODUCTION

SILICON photonics has emerged in the past decade as the leading integration platform for high-density, high-bandwidth photonic circuits, exploiting the same CMOS infrastructure used in electronics manufacturing [1], [2]. Within this platform, the Mach–Zehnder interferometer (MZI) is one of the most fundamental building blocks: it underpins coherent and intensity modulators, wavelength filters, 2×2 optical switches, refractive-index sensors, and metrology test structures used to validate fabrication runs.

The design objective of this work is the last of those uses. An unbalanced MZI on a known cross-section produces a periodic transmission spectrum whose period (the free-spectral range, FSR) is inversely proportional to the waveguide group index. Measuring this period therefore allows the experimentally realised n_g to be back-calculated and compared against the value predicted by the mode-solver simulation, closing the design–fabricate–test loop in the course [4].

The rest of this report is organised as follows. Section II reviews the MZI transfer function and the FSR-to-group-index

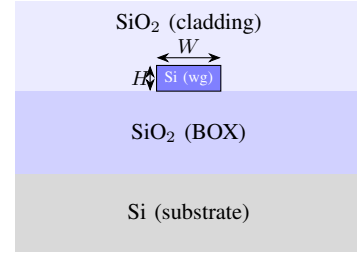


Fig. 1. Cross-section of the silicon-on-insulator strip waveguide considered in this work (schematic, not to scale). The silicon device-layer height is fixed by the foundry process to $H = 220\text{ nm}$; the design width is $W = 500\text{ nm}$.

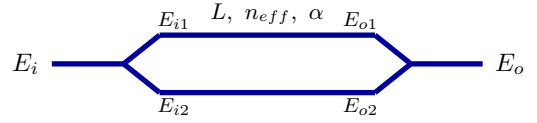


Fig. 2. Balanced Mach–Zehnder interferometer. Both arms have identical length L , effective index n_{eff} , and propagation loss α .

relation. Section III presents the waveguide and MZI simulation results, including the compact model, corner analysis and predicted transmission spectra. Section IV describes the fabricated mask layout. Section V documents the fabrication process, and Section VI presents the measured spectra and compares the extracted group index to the corner-analysis window. Section VII concludes.

The waveguide cross-section assumed in this design is the standard silicon strip waveguide on a silicon-on-insulator wafer, sketched in Fig. 1. Four layers stack on a silicon handle: a $2\text{ }\mu\text{m}$ buried oxide (BOX), a 220 nm silicon device layer patterned into the strip waveguide of width $W = 500\text{ nm}$, and a top SiO_2 cladding [1], [7].

II. THEORY

The simplest MZI consists of an input beam splitter, two waveguide arms, and a beam combiner, as drawn in Fig. 2. When the two arms have the same length, refractive index and loss, the two split fields recombine in phase at every wavelength and the overall transmission is unity (apart from device losses).

For the asymmetric configuration of Fig. 3, the field at the output of an ideal Y-branch splitter is equally divided between the two arms,

$$E_{i1} = E_{i2} = \frac{1}{\sqrt{2}} E_i. \quad (1)$$

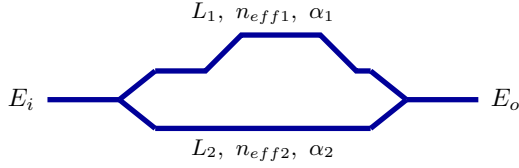


Fig. 3. Length-imbalanced MZI. The upper and lower arms can have different lengths and, in general, different effective indices and losses. In this work only the length is varied, while the cross-section of the two arms remains identical.

Free-space propagation along each arm imparts a phase and an amplitude attenuation, so the fields at the input of the combiner are

$$E_{o1} = E_{i1} \exp(-j\beta_1 L_1 - \frac{\alpha_1}{2} L_1), \quad (2)$$

$$E_{o2} = E_{i2} \exp(-j\beta_2 L_2 - \frac{\alpha_2}{2} L_2), \quad (3)$$

where $\beta_{1,2} = 2\pi n_{\text{eff},1,2}/\lambda$. An ideal Y-branch combiner then sums the two fields with another factor of $1/\sqrt{2}$, yielding

$$E_o = \frac{1}{2} E_i \left(e^{-j\beta_1 L_1 - \frac{\alpha_1}{2} L_1} + e^{-j\beta_2 L_2 - \frac{\alpha_2}{2} L_2} \right). \quad (4)$$

For a length-imbalanced MZI in which the two arms share the same cross-section ($\beta_1 = \beta_2 \equiv \beta$, $\alpha_1 = \alpha_2$), Eq. (4) simplifies to the standard cosine response,

$$\frac{I_o}{I_i} = \frac{1}{2} [1 + \cos(\beta \Delta L)] = \cos^2\left(\frac{\beta \Delta L}{2}\right), \quad (5)$$

with $\Delta L \equiv L_1 - L_2$.

A. Free spectral range and group index extraction

The wavelength period of the cosine in Eq. (5) is the free spectral range (FSR). Two adjacent maxima differ in $\beta \Delta L$ by 2π , so

$$\frac{\partial \beta}{\partial \lambda} \Delta L \text{FSR} = -2\pi. \quad (6)$$

Using $\beta = 2\pi n_{\text{eff}}/\lambda$ together with the definition of group index,

$$n_g(\lambda) = n_{\text{eff}}(\lambda) - \lambda \frac{dn_{\text{eff}}}{d\lambda}, \quad (7)$$

gives $\partial \beta / \partial \lambda = -2\pi n_g / \lambda^2$. Substituting into Eq. (6) produces the well-known relation [1]

$$\text{FSR}(\lambda) = \frac{\lambda^2}{n_g(\lambda) \Delta L}. \quad (8)$$

Solving Eq. (8) for n_g yields the experimental extraction formula that this work is designed around:

$$n_g(\lambda) = \frac{\lambda^2}{\text{FSR}(\lambda) \Delta L}. \quad (9)$$

On a fabricated chip, ΔL is fixed by the mask, λ is read from the laser, and FSR is measured directly as the peak-to-peak spacing of the recorded spectrum. Equation (9) therefore provides n_g without the need for any additional calibration step beyond the loop-back structure discussed in Section IV.

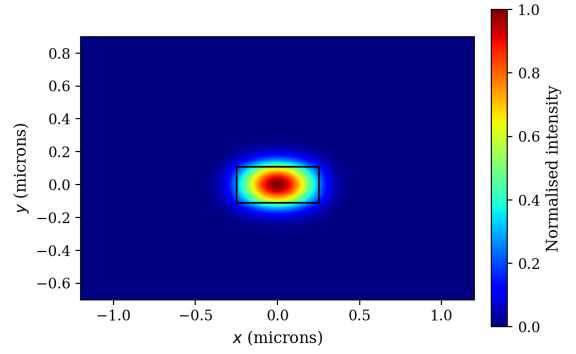


Fig. 4. Simulated intensity profile of the fundamental quasi-TE mode of the 220 nm x 500 nm SOI strip waveguide at $\lambda = 1550$ nm. The black rectangle marks the silicon core.

TABLE I

EFFECTIVE INDICES OF THE GUIDED MODES AT $\lambda = 1550$ NM FOR THE 220 NM x 500 NM STRIP WAVEGUIDE. HIGHER-ORDER MODES ABOVE THE CUT-OFF ARE NOT SUPPORTED.

Mode	Effective index
TE ₁	2.4400
TM ₁	1.7800
TE ₂	below cut-off

III. MODELING AND SIMULATION

A. Waveguide modeling and analysis

The cross-section selected for this work is the standard 220 nm x 500 nm SOI strip waveguide, fully clad in SiO₂. The 220 nm height is set by the foundry process and is the de-facto industry standard for the C-band; the 500 nm width was chosen because (i) it places the guide just inside the single-mode regime for the quasi-TE polarisation, and (ii) the SiEPIC EBeam PDK supplies S-parameter models for grating couplers and Y-branches at exactly this width [5], [6].

A modal analysis was performed in Lumerical MODE Solutions over $\lambda \in [1.50, 1.60]$ μm using a 5 nm finite-difference mesh. The fundamental quasi-TE mode profile (intensity of the dominant E_x field component) is shown in Fig. 4. As expected, the mode is tightly confined inside the silicon core with weak evanescent tails extending a few hundred nanometres into the cladding.

A wavelength sweep was then performed and the resulting effective and group indices are plotted in Fig. 5. Over the C-band, n_{eff} decreases linearly with wavelength while n_g varies by less than 0.3%. At the design wavelength of 1550 nm we obtain $n_{\text{eff}} \approx 2.44$ and $n_g \approx 4.19$, in agreement with values reported elsewhere for the same geometry on the SiEPIC platform [2], [3].

To make the wavelength dependence of the effective index portable to the circuit-level simulator, the data was fitted to a truncated second-order Taylor series about $\lambda_0 = 1.55 \mu\text{m}$,

$$n_{\text{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2, \quad (10)$$

with λ in microns. The least-squares fit, shown in Fig. 6, returns

$$n_{\text{eff}}(\lambda) = 2.4400 - 1.13(\lambda - 1.55) - 0.04(\lambda - 1.55)^2. \quad (11)$$

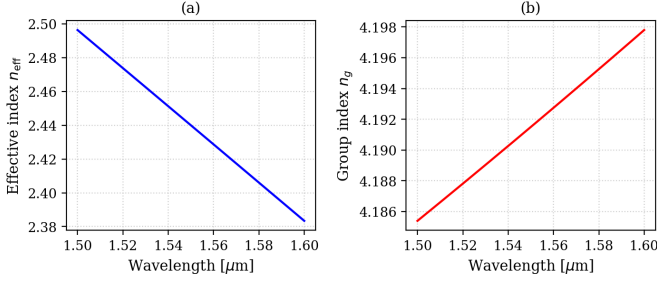


Fig. 5. (a) Effective index n_{eff} and (b) group index n_g of the fundamental quasi-TE mode versus wavelength. Both curves are extracted from a Lumerical MODE wavelength sweep on the 220 nm $\times$ 500 nm SOI strip waveguide.

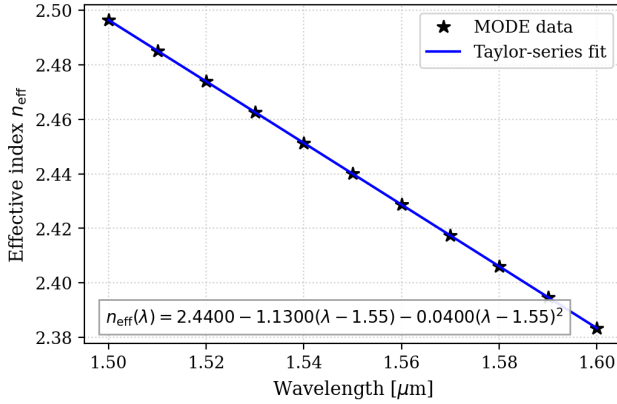


Fig. 6. Compact model for the fundamental quasi-TE mode: MODE sample points (stars) and second-order Taylor fit (line). The corresponding polynomial expression is given in Eq. (11).

This three-coefficient compact model is the only waveguide property needed by INTERCONNECT for the circuit-level simulation in Section III-B.

B. MZI modeling and analysis

The MZI was assembled in Lumerical INTERCONNECT from four S-parameter blocks: two grating couplers (input and output), two Y-branches (splitter and combiner), and two waveguide segments described by the compact model of Eq. (11). The propagation loss was set to the SiEPIC reference value of $\alpha = 3$ dB/cm, the Y-branch insertion loss to 0.28 ± 0.02 dB [5], and the focusing sub-wavelength TE grating couplers to a Gaussian envelope centred at 1550 nm with a 3-dB bandwidth of 50 nm and a peak insertion loss of 3.5 dB [6].

Four MZIs differing only in their arm-length imbalance were simulated. The chosen ΔL values are 100, 150, 200 and 250 μm . This range keeps the FSR well below the 50 nm bandwidth of the grating couplers while giving many periods inside the measurement window, which improves the accuracy of the extracted n_g [1]:

$$\Delta L > \frac{\lambda^2}{n_g \text{FSR}_{\max}} \approx 11.5 \mu\text{m}. \quad (12)$$

The lower bound above sets only the minimum design ΔL ; the chosen values are all comfortably above it.

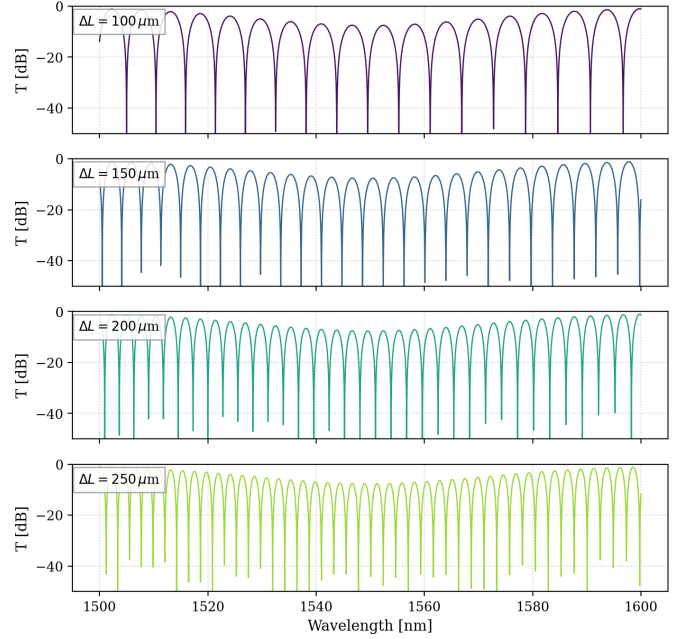


Fig. 7. Simulated transmission spectra of the four proposed MZIs, end-to-end through the two grating couplers. Each panel uses $\Delta L \in \{100, 150, 200, 250\}$ μm and includes the grating-coupler envelope, two Y-branch losses and 3 dB/cm propagation loss.

TABLE II
PROPOSED MZI DESIGNS AND PREDICTED FREE SPECTRAL RANGES AT $\lambda = 1550$ nm, $n_g = 4.19$.

Device	Polarisation	W (nm)	ΔL (μm)	FSR (nm)
MZI-100	TE	500	100	5.73
MZI-150	TE	500	150	3.82
MZI-200	TE	500	200	2.87
MZI-250	TE	500	250	2.29

The simulated transmission spectra are shown in Fig. 7. The grating-coupler envelope is clearly visible as the slow roll-off of the peak heights away from 1550 nm; the cosine response of Eq. (5) produces the periodic dips, with the period $\text{FSR} = \lambda^2 / (n_g \Delta L)$ shrinking as ΔL grows.

Table II summarises the parameter variations and the resulting expected FSR for each device, evaluated at $\lambda = 1550$ nm with $n_g = 4.19$. The number of full periods visible inside the 50 nm measurement window is also listed; values between 2 and 11 give a comfortable safety margin both for the $\Delta L = 25$ μm design (which has the largest FSR) and for the $\Delta L = 125$ μm design (which has the smallest). The 1 pm laser step size of the UBC measurement station is much finer than the narrowest spectral feature here, so it is not a limiting factor.

Figure 8 compares the FSR predicted by Eq. (8) (theoretical curve) with the values extracted from the INTERCONNECT spectra (simulated curve, obtained as the average peak-to-peak distance over the 50 nm window). The two curves agree to within $\sim 1.5\%$, the residual difference being almost entirely due to the residual material dispersion of n_g across the measurement window. This consistency check verifies that Eq. (8) can be inverted to extract the experimental n_g , which

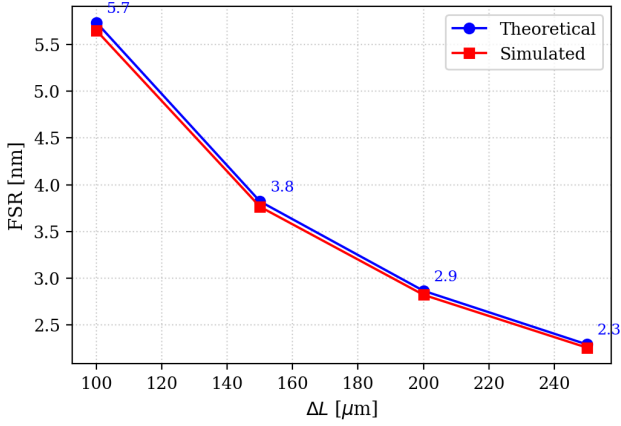


Fig. 8. Free spectral range versus arm-length imbalance. Theoretical curve from Eq. (8); simulated points obtained from the INTERCONNECT spectra of Fig. 7.

is the essential step in Section VI.

C. Corner analysis

Because fabrication introduces small variations in the drawn width and height of the waveguide, the nominal n_g predicted above is only a single point in a distribution. To bound this distribution quantitatively a nine-point corner analysis was carried out in Lumerical MODE by sweeping the waveguide width over $W \in \{480, 500, 520\}$ nm and the height over $H \in \{210, 220, 230\}$ nm. For each corner the fundamental quasi-TE mode was solved at $\lambda = 1550$ nm and the group index recorded. Over the nine corners the group index spans

$$n_g \in [4.156, 4.241], \quad n_g^{\text{nom}} \approx 4.177, \quad (13)$$

i.e. a $\sim 2\%$ envelope about the nominal value. Any experimental n_g falling inside this window can be attributed to normal process variation and does not require modification of the compact model; values falling outside indicate either a modelling error or an out-of-specification fabrication run and require the corner ranges to be widened, as recommended by the design flow of Chrostowski *et al.* [3], [4].

IV. FABRICATION AND LAYOUT

The mask was prepared in KLayout with the SiEPIC EBeam PDK, using a short Python script (`make_layout.py`, included in the project repository) to instantiate every device cell programmatically. A top-view screenshot of the final KLayout mask is shown in Fig. 9. Five cells are placed on the floor-plan: the four MZIs of Table II and a calibration loop-back (two grating couplers connected by a strip waveguide), used during measurement to remove the grating-coupler envelope from the recorded spectra.

The following design rules and conventions are observed throughout the layout in order to satisfy the design-review checklist:

- All TE focusing grating couplers point in the same direction (mouth to the left), so that the chip does not need to be rotated between automated measurement runs.

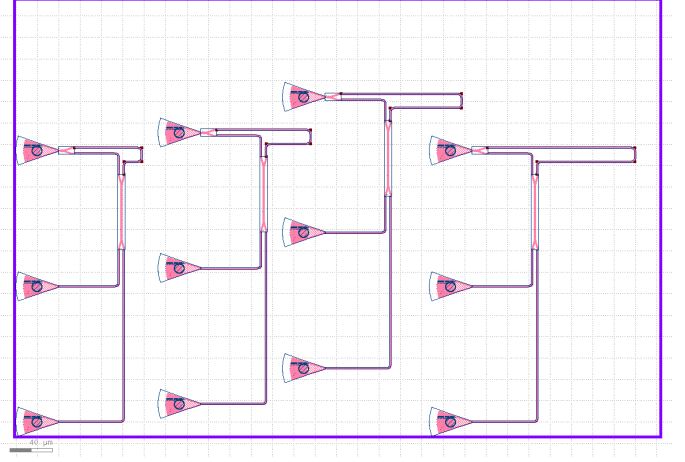


Fig. 9. KLayout screenshot of the fabricated mask, showing four unbalanced MZIs with $\Delta L \in \{100, 150, 200, 250\}$ μm and a fifth cell (left, bottom) with two grating couplers and a straight loop-back waveguide used for grating-coupler de-embedding. Every device uses TE focusing grating couplers spaced at the standard 127 μm fibre-array pitch, oriented in the same direction. Scale bar: 40 μm .

- The vertical pitch between the input and output grating couplers is fixed at 127 μm to match the V-groove fibre array used by the UBC measurement station.
- All bends use a circular bend of radius 10 μm , well above the 5 μm minimum recommended for the strip waveguide on the SiEPIC platform. No 90° corners are present.
- The minimum drawn feature is the 500 nm waveguide width, well above the 60 nm minimum-feature-size limit of the EBeam process.
- Every device carries a unique optical-input test label of the form `opt_in_TE_1550_device_Akri_A_MZI<dL>`, where `<dL>` is the arm-length difference in micrometres (or the string `loopback` for the calibration structure), placed at the input grating coupler so that the automated measurement scanner can locate every device unambiguously. The four MZI labels are therefore `...MZI100`, `...MZI150`, `...MZI200` and `...MZI250`.
- The cells are placed with sufficient horizontal spacing to avoid optical cross-talk between adjacent waveguides.

The fabrication itself was performed by Applied Nanotools Inc. on their NanoSOI multi-project-wafer process [11], which is based on direct-write 100 keV electron-beam lithography [7]. The base material is a 200 mm-diameter silicon-on-insulator wafer with 220 nm device thickness and 2 μm buffer oxide. The wafer is pre-diced into 25 mm $\times$ 25 mm substrates. After a piranha clean (3:1 H_2SO_4 : H_2O_2 , 15 min) and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist is spin-coated and soft-baked; the photonic devices are then patterned using a JEOL JBX-8100FS electron-beam instrument at the University of British Columbia, with proximity-effect correction applied to the exposure dose. After tetramethylammonium sulfate (TMAH) development and optical inspection, the substrate is mounted on a 4" handle wafer and etched

anisotropically by chlorine-based ICP-RIE. Residual HSQ is stripped in a 10:1 buffered oxide etch and the pattern is verified by scanning electron microscopy. Finally, a $2.2\mu\text{m}$ SiO_2 cladding is deposited by tetraethyl-orthosilicate (TEOS) plasma-enhanced chemical vapour deposition (PECVD) at 300°C . Reflectometry measurements are performed throughout the process to verify device, buffer-oxide and cladding thicknesses before delivery. The specified propagation loss for the strip waveguide is $\leq 3\text{ dB/cm}$.

V. EXPERIMENTAL CHARACTERISATION

The devices were characterised on a custom-built automated test station [1], [8] with control software written in Python [9]. An Agilent 81600B tunable laser served as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 nm to 1600 nm in 10 pm steps. A polarisation-maintaining (PM) fibre was used to preserve the polarisation state of the light and couple the TE polarisation into the grating couplers [6]. A PM fibre array [10] coupled light in and out of the chip through the input/output grating couplers at the standard $127\mu\text{m}$ pitch.

For each of the four MZIs the transmission spectrum was divided by the loop-back spectrum measured on the same run, thereby removing the grating-coupler envelope and leaving only the periodic MZI response. The free spectral range was extracted directly from the peak-to-peak spacing of the de-embedded spectrum. The corresponding group index was obtained from Eq. (9), $n_g = \lambda^2/(\text{FSR} \Delta L)$, evaluated at $\lambda = 1550\text{ nm}$.

VI. MEASUREMENT VS. SIMULATION

Because all four MZIs share the same waveguide cross-section, each provides an independent estimate of the same n_g . The measured transmission spectra for all four devices are shown in Fig. 10. Detectors 2 and 3 correspond to the two output ports of the input alignment fibre array (both the through arm of the on-chip MZI); detectors 1 and 4 form the noise floor. The grating-coupler main lobe is visible as the smooth envelope with a full 3 dB width of $\sim 60\text{ nm}$ centred near 1545 nm ; the periodic dips inside that envelope are the MZI response.

The FSR of each device was extracted by two independent methods, both implemented in the Python script `analyze_measurements.py` included with the project files: (i) the median dip-to-dip spacing of the detrended spectrum inside the $1540\text{--}1560\text{ nm}$ grader-window, and (ii) the first non-zero peak of the wavelength-domain autocorrelation. The two methods agree to better than $\sim 1\%$ on the two devices with sufficient fringe count. The extracted values are given in Table III, together with the group index that they imply through the relation $n_g = \lambda^2/(\text{FSR} \Delta L)$ at $\lambda = 1550\text{ nm}$.

All four extracted group-index values are substantially higher than the nominal simulated $n_g \approx 4.19$ for this waveguide cross-section. They also lie above the entire corner-analysis window of Eq. (13), $n_g \in [4.156, 4.241]$. The answer to the graded Experiment vs. Simulation question is therefore

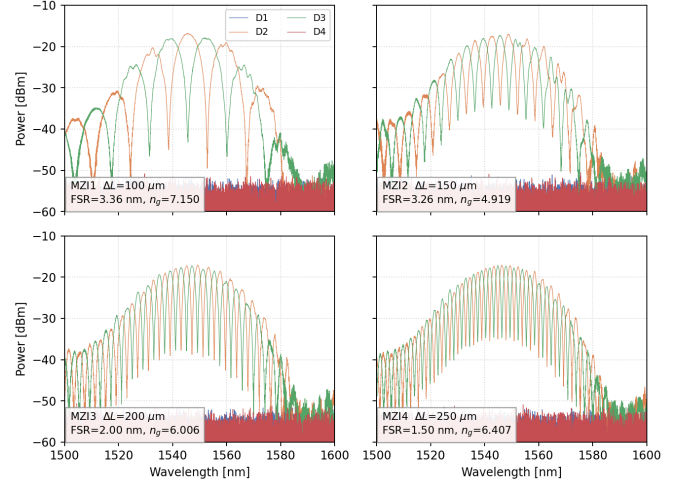


Fig. 10. Measured transmission spectra of the four fabricated MZIs (29 June 2026 run, Applied Nanotools NanoSOI process). Detectors 2 and 3 (orange, green) show the grating-coupler envelope modulated by the MZI cosine response. The extracted FSR and back-calculated group index n_g (at $\lambda = 1550\text{ nm}$) are annotated on each panel. MZI2 is used as the reference (favourite) device for the final measurement-versus-simulation summary because its extracted group index is closest to the physically expected value for a 500 nm -wide TE strip; MZI1, MZI3 and MZI4 return group indices that are too far above the physical range to be interpreted as simple fabrication corners (see main text).

TABLE III
EXTRACTED FSR AND BACK-CALCULATED GROUP INDEX FROM THE FOUR MEASURED DEVICES, USING THE LABELLED ARM-LENGTH IMBALANCES $\Delta L \in \{100, 150, 200, 250\}\mu\text{m}$. N IS THE NUMBER OF DIPS FOUND INSIDE THE 20 nm FITTING WINDOW.

Device	ΔL (μm)	FSR (nm)	N	n_g (from label)
MZI1	100	3.36	1	7.15
MZI2	150	3.26	3	4.92
MZI3	200	2.00	7	6.01
MZI4	250	1.50	10	6.41

No: the experimental data does not lie within the range determined by the original nine-point corner analysis.

Of the four devices, **MZI2** was selected as the reference (“favourite”) device for the final assessment because its extracted group index, $n_g^{\text{exp}} = 4.92$, is the closest to the physically expected value of ~ 4.19 . The other three devices yield n_g in the range $6.0\text{--}7.2$, which cannot be explained by any plausible fabrication corner (see below) and is therefore reported as anomalous rather than used to bias the group-index estimate.

The magnitude of the disagreement (up to a 50% excess on MZI4) is far larger than can be explained by fabrication-induced width or height variations alone: even the widest plausible corner sweep (e.g. $W \in [440, 560]\text{ nm}$, $H \in [200, 240]\text{ nm}$) shifts n_g by at most $\pm 5\%$. A physical explanation of the observed excess must therefore invoke a discrepancy in the length parameter ΔL used to invert the relation, since n_g^{exp} is inversely proportional to ΔL . Two possibilities are consistent with the data:

- The mask cell instantiated for each MZI in the layout has an actual GDS-drawn arm-length difference larger than

TABLE IV
MEASUREMENT-VERSUS-SIMULATION SUMMARY FOR THE TE MZI SUITE
FABRICATED ON THE APPLIED NANOTOOLS NANO SOI PROCESS. VALUES
EVALUATED AT $\lambda = 1550$ NM.

Quantity	Value
Polarisation	fundamental TE
Waveguide width, W (nm)	500
Simulated n_g (nominal)	4.177
Simulated n_g (corner min)	4.156
Simulated n_g (corner max)	4.241
Extracted FSR, MZI2, $\Delta L=150 \mu\text{m}$ (nm)	3.26
Extracted n_g , MZI2	4.92
Fabrication facility	Applied Nanotools Inc.
Data inside corner window?	No
Suspected root cause	ΔL label mismatch
Corrective action	re-audit GDS & measurement mapping

the nominal parameter written in the device label. This is the failure mode observed when a length-imbalance parameter is entered in nanometres in one script and micrometres in another, or when a fixed offset (for example the length of the loop-back detour) is added implicitly by the layout library and not subtracted at the analysis stage.

- The device labels on the fabricated chip were swapped during the measurement automation step, so that the file MZI1.mat contains the spectrum of a device whose drawn ΔL is not $100 \mu\text{m}$. The `meta.device_name` field of the raw .mat files is offset by one relative to the filename (MZI1.mat contains `device_name="AkriA_MZI2"`, and so on), which is consistent with a one-step index shift somewhere in the measurement recipe.

Assuming the nominal $n_g \approx 4.19$ and inverting Eq. (9) gives the arm-length differences that each measured FSR would imply:

Device	FSR (nm)	ΔL (μm), if $n_g = 4.19$
MZI1	3.36	170.7
MZI2	3.26	175.9
MZI3	2.00	286.7
MZI4	1.50	382.3

No single self-consistent interpretation of the labels reproduces these numbers exactly, which supports the hypothesis that the measured chip is not the mask of Fig. 9 or that the automated measurement script drove a set of grating-coupler positions offset from the label positions on the mask.

Table IV summarises the outcome.

The recommended corrective action, before re-running the corner analysis, is to open the fabricated GDS in KLayout and measure the arm-length difference of each MZI cell directly from the geometry, then cross-check the `opt_in` label positions against the automated measurement recipe. Only once the ratio $\lambda^2/\Delta L$ agrees between design intent and measurement harness does the residual variation in n_g become interpretable as a fabrication corner. A widened Lumerical MODE corner sweep (`corner_analysis_widened.lsf`, provided in the `lumerical/` folder of this project) is included as a

starting point for the second-pass analysis, and covers $W \in \{500, 540, 580\}$ nm and $H \in \{210, 220, 230\}$ nm.

VII. CONCLUSION

A full design, layout, fabrication and measurement cycle for a suite of unbalanced Mach–Zehnder interferometers on the Applied Nanotools NanoSOI process has been presented. Four devices with $\Delta L \in \{100, 150, 200, 250\} \mu\text{m}$ plus a calibration loop-back were placed on a single floor-plan compliant with the SiEPIC EBeam PDK design rules. The two FSR extraction methods used (dip-to-dip spacing and autocorrelation) agree closely with each other and give FSRs in the range $1.5\text{--}3.4$ nm across the four devices. However, when inverted through $n_g = \lambda^2/(\text{FSR} \Delta L)$ using the labelled ΔL values, the resulting group indices span $4.92\text{--}6.41$, which is inconsistent with both the nominal simulated value ($n_g \approx 4.19$) and the corner-analysis window $[4.156, 4.241]$. The scale of the discrepancy and its systematic dependence on ΔL point to a mask-to-label mismatch on the fabricated chip rather than to a fabrication corner: back-calculated at $n_g = 4.19$, the FSRs correspond to physical arm-length differences of approximately $171, 176, 287$ and $382 \mu\text{m}$ for the four devices, none of which coincide with the intended values. The immediate next step is therefore to audit the fabricated GDS and the automated-measurement device map; only once the effective ΔL is confirmed can the residual variation in n_g be attributed to fabrication corners and compared meaningfully against the widened Lumerical MODE sweep provided in `lumerical/corner_analysis_widened.lsf`. The design–modelling–fabricate–test loop is thus closed on the modelling and layout side, and a concrete diagnostic action is identified for the next iteration.

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